24ME311 PRODUCT DEVELOPMENT LAB-1

FAKE JOB POSTING DETECTION SYSTEM

A PROJECT REPORT

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ABSTRACT

Online recruitment platforms are increasingly targeted by fraudulent entities that publish fake job advertisements to collect personal information, application fees, or unpaid work. Traditional detection systems often rely mainly on text features, provide limited explanations, and may struggle with multilingual or code-mixed postings. This project proposes the SecureHire Framework, a hybrid Fake Job Posting Detection System that combines Artificial Intelligence, Machine Learning, Natural Language Processing, rule-based checks, and company verification. Job postings from public datasets such as EMSCAD and Kaggle are cleaned and transformed using TF-IDF, Bag-of-Words, or contextual embeddings, then classified using models such as Random Forest, Artificial Neural Network, mBERT, and XLM-RoBERTa. The system also checks suspicious indicators such as free email domains, WhatsApp-only contact details, unrealistic salaries, and urgency phrases, while planned verification checks compare company information with relevant official records. SHAP and LIME provide explanations for predictions, and a Streamlit interface presents a verdict, risk score, verification status, and apply-or-avoid guidance. The report describes the problem, existing and proposed systems, literature survey, system architecture, preprocessing steps, evaluation metrics, limitations, and future enhancement.

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LIST OF ABBREVIATIONS

S. No.	ABBREVIATION	EXPANSION
1	AI	Artificial Intelligence
2	ANN	Artificial Neural Network
3	API	Application Programming Interface
4	CNN	Convolutional Neural Network
5	EDA	Exploratory Data Analysis
6	EMSCAD	Employment Scam Aegean Dataset
7	FTC	Federal Trade Commission
8	GST	Goods and Services Tax
9	LIME	Local Interpretable Model-agnostic Explanations
10	LSTM	Long Short-Term Memory
11	MCA	Ministry of Corporate Affairs
12	mBERT	Multilingual Bidirectional Encoder Representations from Transformers
13	ML	Machine Learning
14	NLP	Natural Language Processing
15	SHAP	SHapley Additive exPlanations
16	SMOTE	Synthetic Minority Over-sampling Technique
17	SVM	Support Vector Machine
18	TF-IDF	Term Frequency – Inverse Document Frequency
19	UI	User Interface
20	UIDAI	Unique Identification Authority of India
21	XAI	Explainable Artificial Intelligence
22	XLM-RoBERTa	Cross-lingual Language Model – Robustly Optimized BERT Approach

CHAPTER 1

INTRODUCTION

1.1 Problem Statement

The internet has made job hunting faster, but it has also given fraudsters an easy way to reach job seekers. Fake job postings imitate genuine openings to collect personal data, application fees or unpaid work. The financial impact is large: Americans lost about US $450 million to fake job scams in 2024 (FTC), Australia reported a total loss of AU $2.74 billion, and Australian job-scam losses rose by 150% compared with 2022. In the EMSCAD dataset, 12% of the sampled Workable postings were fraudulent.

Automated detection matters now for four reasons. Manual review cannot scale to thousands of daily postings across global platforms. Scams disproportionately target vulnerable groups such as students, entry-level applicants and flexible-work seekers. Fraudulent postings mimic legitimate ones closely, which defeats simple keyword filters. Finally, an accurate, explainable and multilingual detection system builds trust in online recruitment.The problem addressed in this project is therefore to build a detection system that does not rely on text analysis alone, that explains why a posting is flagged, that verifies the recruiting company against official records, and that works for multilingual and code-mixed posting.

1.2 Objectives

- To collect job postings from public datasets (EMSCAD, Kaggle Balanced Fake Job Postings) and live recruitment platforms across multiple languages.
- To clean and normalise the text and convert it into numerical features (TF-IDF, Bag-of-Words, contextual embeddings).
- To classify each posting as Real or Fake using Random Forest, ANN, mBERT

and XLM-RoBERTa models.

- To add rule-based checks and government / external verification (MCA, GST, Companies House, SEC EDGAR, phone and domain checks) so that the company's authenticity is confirmed.
- To explain every verdict with SHAP (global feature importance) and LIME (word-level explanation of individual predictions).
- To present the verdict, risk level and apply-or-avoid guidance to the user through a web interface.
- To use hierarchical clustering and dashboards to reveal scam sub-types and geographic hotspots.

1.2 Scope of the Project

The project covers the design of the SecureHire Framework: a pipeline that runs from data collection to a final fraud verdict. It includes eight modules (data collection, preprocessing, feature extraction, visualization and clustering, classification, rule-based/ government / external verification, explainability, and recommendation with UI). The system targets multilingual and code-mixed job postings, uses Python-based machine-learning and transformer libraries, and is delivered as a Streamlit / Flask web application. Government verification is planned against Indian (MCA, GST, UIDAI), UK (Companies House) and US (SEC EDGAR) sources.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

Existing fake-job detection systems fall into the following groups:

- Classical machine learning (Naive Bayes, SVM, Logistic Regression) on TF-IDF text features.
- Ensemble models (Random Forest, AdaBoost, Gradient Boosting) used for higher accuracy.
- Deep learning models such as Bi-LSTM, GRU and attention-based sequence models.
- Single-language transformer or classification pipelines deployed as web applications (for example, Streamlit).
- Exploratory data analysis and clustering, used mainly for descriptive insight rather than real-time prediction.

Key limitations of the existing systems

- They function largely as “black boxes”, giving no explanation for why a posting is flagged.
- They rely on text analysis only, with no verification of recruiter identity or company legitimacy.
- Most are trained on a single, imbalanced, English-only dataset (EMSCAD).
- They struggle with multilingual and code-mixed job postings written in regional languages.
- High-accuracy deep-learning and transformer models demand heavy compute and frequent retraining.

2.2 PROPOSED SYSTEM

The proposed system is the SecureHire Framework, an AI + NLP + Government-Verification approach to fake job posting detection. It combines NLP and machine learning with multi-layer verification, including government verification, to analyse job postings and decide whether they are genuine or fake.

Features to be implemented

- **Hybrid Verification:** verifies company registration and domain.
- **Real-Time NLP Filter:** detects scam keywords in job descriptions.
- **Behavioral Tracker:** identifies interview-farming patterns.
- **IP & Assignment Guard:** detects suspicious unpaid-work requests.
- **Auto Quarantine:** isolates and flags high-risk job postings.

Government verification (new step): the system cross-checks company details (registration, GST, domain, licence) with official databases such as MCA (company registration), GST (tax details) and UIDAI (identity / business KYC). The step validates company details, verifies domain and e-mail, checks licence / registration and detects fake identities.

Processing flow: Dataset → Data Preprocessing (missing values, text cleaning, normalisation, tokenisation) → Feature Extraction (TF-IDF / Bag-of-Words / contextual embeddings) → Deep Learning / ML Model → Government Verification → Anomaly Detection (suspicious patterns, high-risk flags) → Fraud Prediction (Genuine / Fake). A

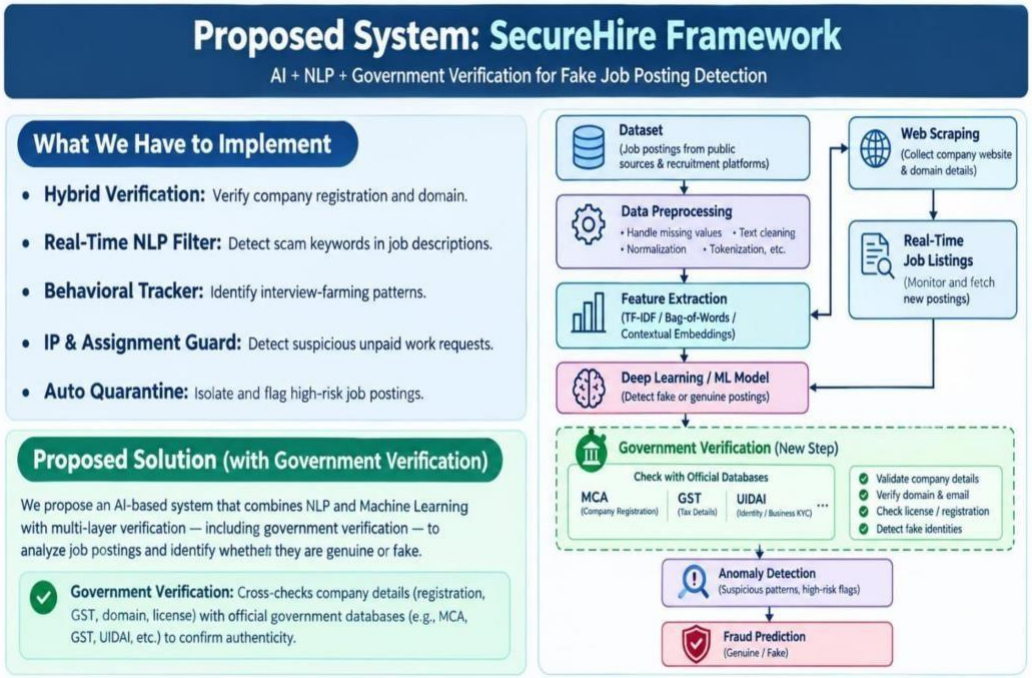


Table 2.1 Comparison of existing systems and the proposed system

Aspect	Existing systems	Proposed SecureHire Framework
Analysis basis	Text analysis only	Text + rule-based checks + government and external verification

Aspect	Existing systems	Proposed SecureHire Framework
Explainability	"Black box"	SHAP and LIME explanations
Language	Single-language,	Multilingual and code-mixed

support	English-only	(mBERT, XLM-RoBERTa)
Company legitimacy	Not verified	Checked against MCA / GST / UIDAI, Companies House, SEC EDGAR
Data	Single imbalanced dataset	Public datasets plus live postings
Output	Real / Fake label	Real / Medium-Risk / Fake verdict, risk score, apply-or-avoid guidance

2.3 SYSTEM REQUIREMENTS SPECIFICATION

2.3.1. Software Requirements

- Language: Python 3.9+
- ML / DL: scikit-learn, TensorFlow / Keras, PyTorch, HuggingFace Transformers (mBERT, XLM-RoBERTa)
- NLP: NLTK / spaCy for tokenisation, lemmatisation and stop-word filtering
- Explainability: SHAP, LIME
- Visualization / EDA: Tableau, Orange Data Mining, Matplotlib / Seaborn
- Deployment: Streamlit / Flask web app
- External and government APIs: OpenCage, Twilio, OpenCorporates, WhoisXML, MCA21 / GST (India), Companies House (UK), SEC EDGAR (US)
- Database: MySQL / MongoDB for postings and prediction logs

2.3.2. Hardware Requirements

- Processor: Intel i5 / AMD Ryzen 5 or higher (i7 recommended for transformer

training)

- RAM: 8 GB minimum; 16 GB or more recommended for transformer fine-tuning
- GPU: CUDA-enabled GPU (for example, NVIDIA RTX series) recommended for mBERT / XLM-RoBERTa training
- Storage: 20 GB or more of free space for datasets, embeddings and model checkpoints
- Operating system: Windows 10/11, Linux (Ubuntu) or macOS
- Network: Internet connectivity for external API verification calls
- Browser: Chrome / Edge / Firefox for the Streamlit web interface

CHAPTER 3 LITERATURE SURVEY

3.1 REVIEW

The papers below were studied to understand the methods, outcomes and evaluation metrics used for fake job detection. Earlier work uses classical classifiers on linguistic and structural features; later work moves to deep learning, class-imbalance handling, hybrid rule-based approaches and multilingual, explainable transformers.

Table 3.1 Existing solutions for fake job detection

Paper / Source	Year	Methodology applied	Core outcome achieved	Evaluation metrics
Vidros et al.	2017	Random Forest + Bag-of-Words	Detected fake job postings using linguistic and structural patterns	Accuracy, Precision, Recall, F1-Score
Alghamdi et al.	2019	SVM + Random Forest	Improved detection using company profile, logo, industry and job details	Accuracy, Precision, Recall
Lai et al.	2019	Ensemble Learning / ORF	Improved detection using linguistic, contextual and metadata features	Accuracy, Precision, Recall, F1-Score
Dutta Bandyopadhya	2020	Decision Tree, Random Forest, MLP, KNN	Decision Tree and Random Forest achieved better classification performance	Accuracy, Precision, Recall

Ranparia et al.	2020	NLP + GloVe + Sequential Neural Network	High detection performance on LinkedIn job postings	Accuracy, Precision, Recall

Table 3.2 Advanced models and hybrid approaches

Paper / Source	Year	Methodology applied	Core outcome achieved	Evaluation metrics
Habiba et al.	2021	Supervised ML + Deep Neural Network	High classification accuracy for identifying scam job posts	Accuracy, ROC-AUC, False Positive Rate
Vo et al.	2021	TF-IDF + SVM + SMOTE + Logistic Regression	Improved classification after handling class imbalance	Accuracy, Precision, Recall, F1-Score

Paper / Source	Year	Methodology applied	Core outcome achieved	Evaluation metrics
Dalish et al.	2022	Convolutional Neural Network (CNN)	CNN outperformed several traditional ML methods	Accuracy, Precision, Recall
Shelar et al.	2026	ML + NLP + Rule-based + External Verification	Combined job, recruiter, salary and e-mail features for improved scam detection	Accuracy, Precision, Recall, F1-Score
Nivetha et al.	2026	mBERT XLM-RoBERTa +	Improved multilingual detection with	Accuracy, Precision,Rec

		SHAP/LIME	explainable predictions	all, F1-Score

Observations from the survey

- Tree-based and ensemble models (Random Forest, Decision Tree) perform consistently well on structured and linguistic features.
- Handling class imbalance (for example, with SMOTE) improves classification, which matters because fake postings are a small minority in EMSCAD.
- Deep models (CNN, sequential networks, deep neural networks) outperform several traditional methods, but at a higher compute cost.
- Recent work adds rule-based checks, external verification, multilingual transformers and explainability. The SecureHire Framework combines all of these in a single pipeline.

CHAPTER 4 SYSTEM ARCHITECTURE

4.1 Block Diagram

The system is organised as seven layers, followed by the final fraud-detection result: Data Layer → Preprocessing Layer → Feature Extraction Layer → Machine Learning Layer → Rule-Based & Verification Layer → Explainability & Evaluation Layer → Recommendation & UI Layer → Fraud Detection Result.

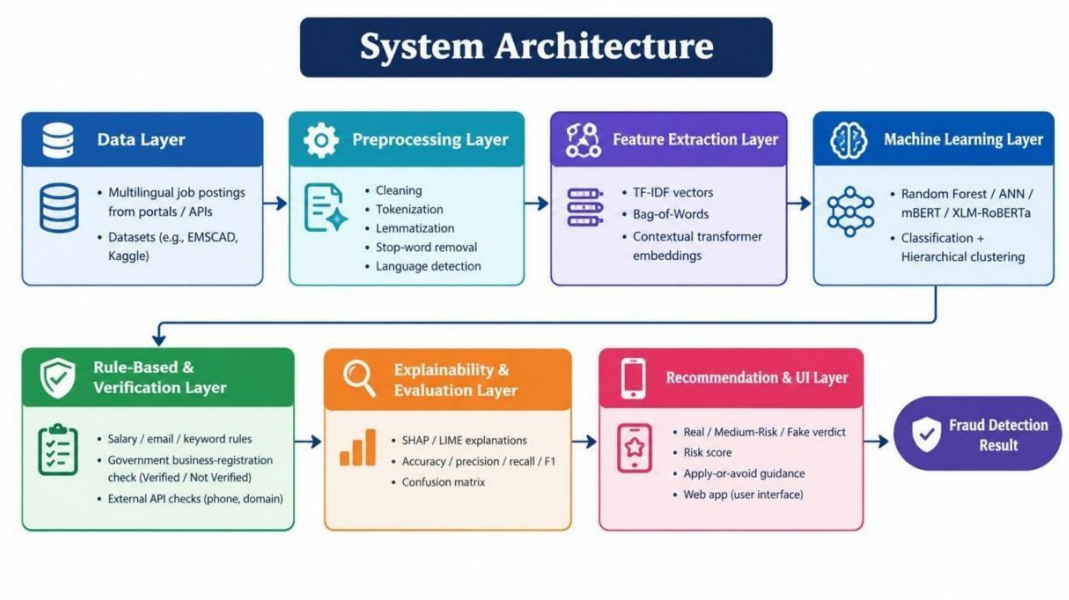


Fig: 4.1.a System Architecture Diagram

4.2.Diagram Description

- **Data Layer:** collects multilingual job postings from portals and APIs and from datasets such as EMSCAD and Kaggle.
- **Preprocessing Layer:** performs cleaning, tokenisation, lemmatisation, stop-word removal and language detection.
- **Feature Extraction Layer:** produces TF-IDF vectors, Bag-of-Words features and contextual transformer embeddings.
- **Machine Learning Layer:** applies Random Forest, ANN, mBERT

and XLM-RoBERTa for classification, and hierarchical clustering to reveal scam sub-types.

- **Rule-Based & Verification Layer:** applies salary, e-mail and keyword rules; performs the government business-registration check (Verified / Not Verified); and makes
- external API checks on phone numbers and domains.
- **Explainability & Evaluation Layer:** produces SHAP / LIME explanations and computes accuracy, precision, recall, F1-score and the confusion matrix.
- **Recommendation & UI Layer:** shows the Real / Medium-Risk / Fake verdict, the risk score and apply-or-avoid guidance through the web application.

Module definitions

Table 4.1 Modules of the SecureHire Framework

No.	Module	Description
1	Data Collection Module	Gathers job postings from public datasets (EMSCAD, Kaggle Balanced Fake Job Postings) and live recruitment platforms across multiple languages.
2	Preprocessing Module	Removes HTML, URLs, punctuation and duplicates; performs lowercasing, tokenisation, stop-word removal and lemmatisation; detects the posting language.
3	Feature Extraction	Converts cleaned text into numerical form using TF-IDF / Bag-of-Words for classical

	Module	models and contextual embeddings for transformer models.
4	Visualization & Clustering Module	Builds Tableau / Orange dashboards (pie, bar, choropleth, word cloud) and performs hierarchical clustering to reveal scam sub-types and geographic hotspots.
5	Classification Module	Random Forest / ANN / mBERT / XLM-RoBERTa classify each posting as Real or Fake using the extracted features; models are trained and evaluated with an 80/20 split.
6	Rule-Based, Government & External Verification Module	Flags free e-mail domains, WhatsApp-only contact, unrealistic salary and urgency phrases; verifies government registration (MCA / GST / Companies House / SEC EDGAR); cross-checks phone and domain via external APIs.
7	Explainability (XAI) Module	SHAP gives global feature importance; LIME explains individual predictions by highlighting the words responsible for the Real / Fake verdict.
8	Recommendation & UI Module	Displays the final verdict (Real / Medium-Risk / Fake), fraud risk level, government verification status (Verified / Not Verified), suspicious features found, and apply / avoid guidance through a Streamlit web interface.

4.2 Preprocessing Steps

Preprocessing prepares postings from different sources and languages so that the feature extractor and the classifiers can treat them uniformly

- **Handling missing values:** incomplete fields in a posting are handled before further processing.
- **Text cleaning:** HTML tags, URLs, punctuation and duplicate postings are removed.
- **Normalisation:** text is converted to lowercase and normalised.
- **Tokenisation:** each posting is split into tokens using NLTK / spaCy.
- **Stop-word removal:** common words that carry little meaning are filtered out.
- **Lemmatisation:** words are reduced to their base forms.
- **Language detection:** the language of each posting is detected so that multilingual and code-mixed postings can be routed to suitable models.
- **Feature extraction:** the cleaned text is converted into TF-IDF vectors, Bag-of-Words features or contextual embeddings.
- **Dataset split:** the data is divided in an 80/20 ratio for training and evaluation.

4.3 Performance Metric

The system is evaluated with the standard classification metrics listed in the Explainability & Evaluation Layer:

Accuracy: the proportion of postings classified correctly.

Precision: the fraction of postings flagged as fake that are actually fake.

Recall: the fraction of actual fake postings that are detected.

F1-score: the harmonic mean of precision and recall, useful because fake postings form a minority class.

Confusion matrix: the counts of true / false positives and negatives for each model.

In addition, SHAP and LIME are used to check that the words and features driving a verdict are sensible, and the government-verification status (Verified / Not Verified) is reported alongside every prediction. The models (Random Forest, ANN, mBERT, XLM-RoBERTa) will be compared on the same 80/20 split, and the results will be recorded in this section once training is complete.

APTER 5 CONCLUSION

This project proposes the SecureHire Framework, a hybrid fake job posting detection system that combines NLP, machine learning, rule-based checks and government verification. It addresses the main weaknesses of existing systems: it adds verification of company identity beyond text analysis, supports multilingual and code-mixed postings through mBERT and XLM-RoBERTa, and explains each verdict through SHAP and LIME. The final output (a Real / Medium-Risk / Fake verdict with a risk score and apply-or-avoid guidance) helps job seekers, especially students and entry-level applicants, avoid fraudulent offers, while features such as real-time filtering and auto-quarantine help platforms act on high-risk postings.

5.1 Limitations

- The work is at the design stage: the architecture, modules and metrics are defined, but experimental results are yet to be recorded.
- Most public data is English-only and imbalanced (EMSCAD); multilingual and code-mixed data must be collected and labelled.
- Government and external verification depends on the availability and access terms of the MCA / GST, Companies House, SEC EDGAR and other APIs, and needs internet connectivity.
- Transformer models (mBERT, XLM-RoBERTa) require heavy compute, a CUDA-enabled GPU and frequent retraining.

5.2 Future Enhancements

- Extend language coverage to more regional languages and code-mixed text.
- Expand verification to additional national company registries and identity sources.
- Strengthen the real-time monitoring of live listings, the Behavioral Tracker (interview-farming patterns) and the IP & Assignment Guard.
- Use clustering and dashboards to track new scam sub-types and geographic hotspots over time.
- Run formal evaluations of accuracy, precision, recall and F1-score on larger dataset

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APPENDIX I – SOURCE CODE

The application “TrueHire – Job Scam Detector” is a single HTML file that runs entirely in the browser. The listing is divided into the application logic (JavaScript) and the interface markup and style sheet (HTML/CSS). Hindi text strings in the listing are written as JavaScript Unicode escapes (\uXXXX) so that they print correctly; they are equivalent to the original characters.

A. Application logic (JavaScript)

```
const FREE = ["gmail.com", "yahoo.com", "outlook.com"];

function analyzeJob(text) {
  const flags = [];
  const email = text.match(/[\w.+-]+@[\w.-]+\.\w+/)?.[0] || "";
  const domain = email.split("@")[1] || "";

  if (/registration fee|security fee|pay.*fee|₹\s*\d+/i.test(text))
    flags.push("Advance fee requested");

  if (/PAN|Aadhaar|bank details|OTP|UPI PIN/i.test(text))
    flags.push("Sensitive information requested");

  if (/no experience|10th pass|work from home/i.test(text) &&
      /₹\s*\d+\s*(daily|per day)/i.test(text))
    flags.push("Unrealistic salary claim");

  if (/no interview|direct joining|bina interview/i.test(text))
    flags.push("No interview promised");

  if (FREE.includes(domain))
    flags.push("Personal email address");
```

```
  const score = Math.min(100, flags.length * 25);

  return {
    score,
    risk: score >= 60 ? "High" :
          score >= 30 ? "Medium" : "Low",
    flags,
    recommendation: score >= 60
      ? "Avoid this job"
      : score >= 30
      ? "Verify before applying"
      : "No major warning signs detected"
  };
}

console.log(analyzeJob(
  "Work from home! Earn ₹3000 daily. Pay ₹499 registration fee."
));
```

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>TrueHire - Job Scam Detector</title>
<style>
body{font-family:Arial;background:#0a0e17;color:white;padding:20px;max-
width:800px;margin:auto}
header,h1{text-align:center;color:#3b82f6}
.card{background:#101726;padding:20px;border-radius:15px;margin:15px 0}
textarea{width:96%;height:180px;padding:10px;background:#0b111c;color:white;b
order:1px solid #334155;border-radius:10px}
button{padding:12px;background:#3b82f6;color:white;border:0;border-
radius:8px;cursor:pointer;margin-top:10px}
.result{padding:15px;border-radius:10px;background:#1c2538}
</style>
</head>
<body>
<header>
<h2>□ TrueHire</h2>
<p>JOB SCAM DETECTOR</p>
</header>
<h1>Real Job or Scam?</h1>
<p>Check job postings in English, Hindi or Hinglish.</p>

<div class="card">
<textarea id="job" placeholder="Paste job posting here..."></textarea><br>
<button onclick="checkJob()">Analyse Posting</button>
</div>

<div class="card result" id="result" hidden></div>
```

```
<script>
function checkJob(){
 const text=document.getElementById("job").value.toLowerCase();
 const flags=[];

 if(/registration fee|security fee|pay.*fee|₹\s*\d+.*fee/i.test(text))
 flags.push("Advance registration fee requested");

 if(/pan card|aadhaar|bank details|otp|upi pin/i.test(text))
 flags.push("Sensitive information requested");

 if(/no experience|10th pass|work from home/i.test(text)&&
 /₹\s*\d+.*(daily|per day)/i.test(text))
 flags.push("Unrealistic salary claim");

 if(/no interview|direct joining|bina interview/i.test(text))
 flags.push("Suspicious direct selection promise");

 if(/gmail\.com|yahoo\.com|outlook\.com/i.test(text))
 flags.push("Personal email address used");

 const score=Math.min(100,flags.length*25);
 const risk=score>=60?"HIGH RISK":
 score>=30?"MEDIUM RISK":"LOW RISK";

 const box=document.getElementById("result");
 box.hidden=false;
 box.innerHTML=`<h2>${risk}</h2>
 <h3>Risk Score: ${score}/100</h3>
 <p>${flags.length?flags.join("<br>"):"No common warning signs detected."}</p>
 <b>${score>=60?"Avoid this job and do not pay.":
 score>=30?"Verify the employer before applying.":
```

```
 "Check the official careers website before applying."}</b>`;
}
</script>
</body>


</html>
```

APPENDIX II – SCREENSHOTS

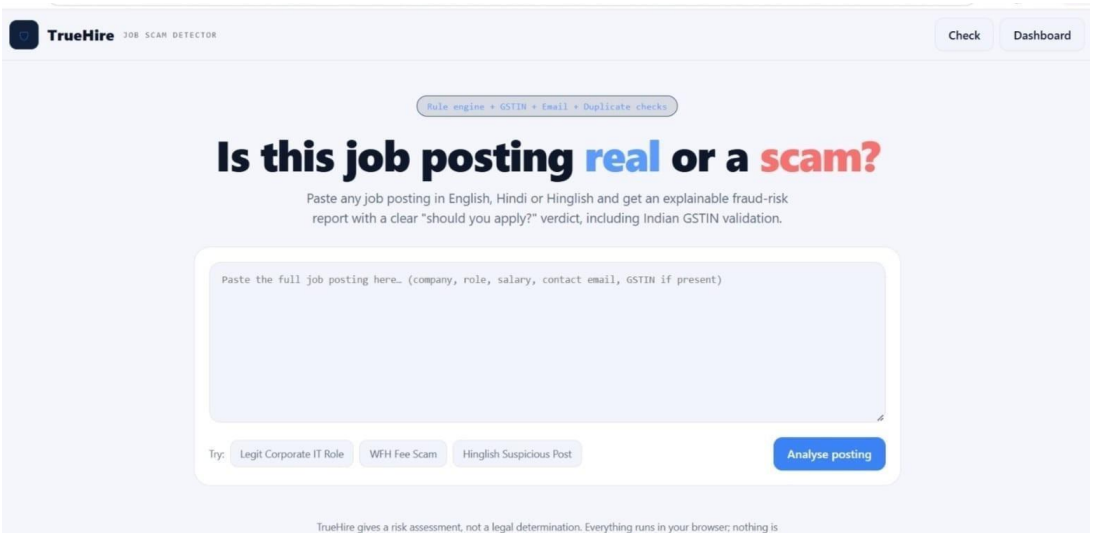


Fig: A.1 TrueHire home screen with job-posting input

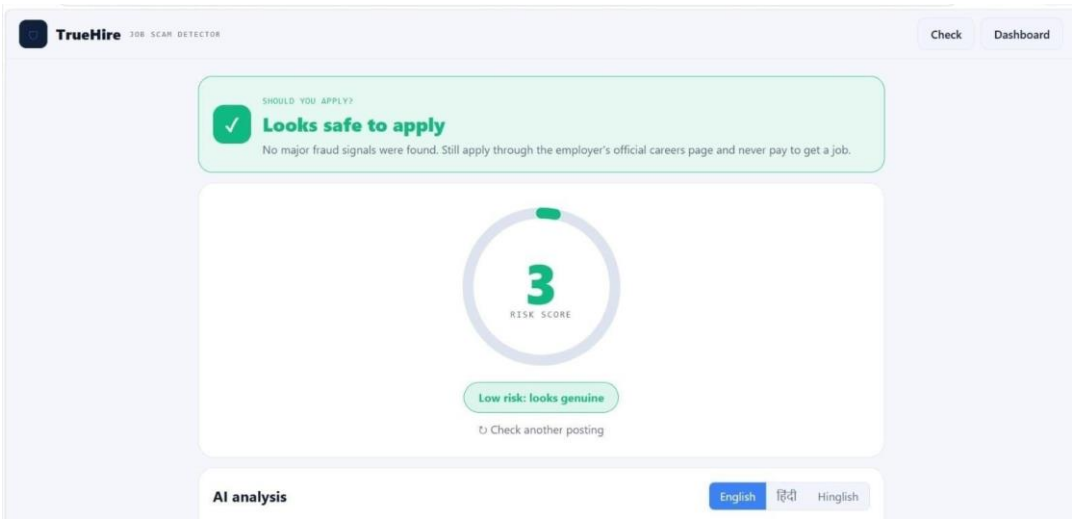


Fig: A.2 Verdict and risk score for a genuine job posting

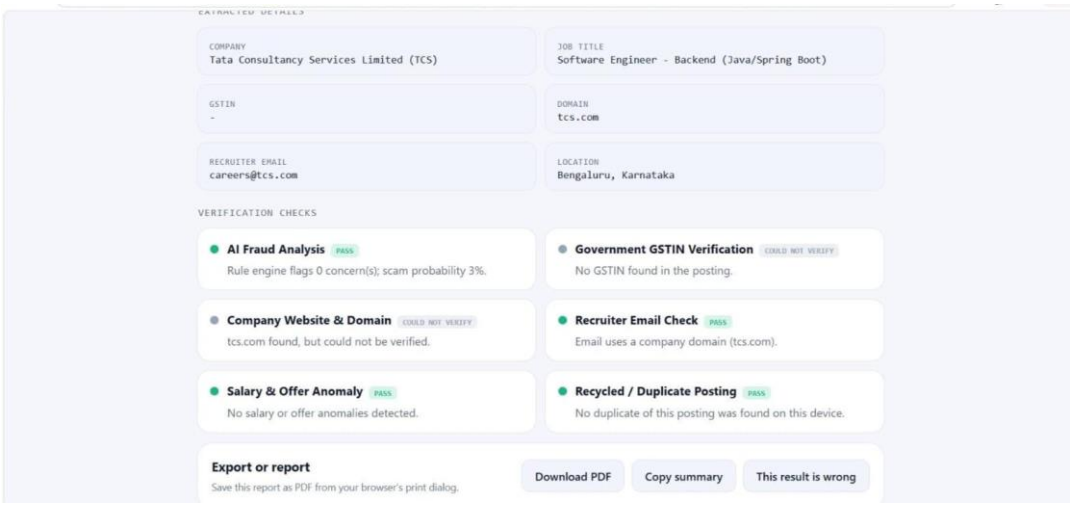


Fig: A.3 Extracted details and verification checks